$$= \sigma^2 + \mu^2 - \frac{\sigma^2}{k} - \mu^2$$

$$= \sigma^2 - \frac{\sigma^2}{k} = \frac{k\sigma^2 - \sigma^2}{k} = \sigma^2\left(\frac{k-1}{k}\right)$$

$$\sigma^2 - \frac{\sigma^2}{k} - \sigma^2 = -\frac{\sigma^2}{k}$$